Derivatives

Original CC0 evaluation fixture - generated locally

The power rule gives d(x^n)/dx = n*x^(n-1).

The product rule gives (fg)' = f'g + fg'. The chain rule gives d f(g(x))/dx = f'(g(x))*g'(x).

Integrals

Original CC0 evaluation fixture - generated locally

For n not equal to -1, an antiderivative of x^n is x^(n+1)/(n+1) plus a constant.

The fundamental theorem says that if F'=f, then the integral of f from a to b is F(b)-F(a).